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Demand Shocks as Technology Shocks

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1 Big picture

- **Core question.** Why does measured TFP (the Solow residual) move so much over the business cycle, and why is it so procyclical, even when many narratives emphasize *demand* fluctuations?
- **Key mechanism.** In many sectors, output is realized only if *customers and producers meet*. A product-market search friction implies that part of “capacity” is idle when shoppers search less. This generates an endogenous utilization margin that is *not recorded* as an input in national accounts.
- **Main claim.** Shocks that raise demand by increasing shopping/search effort (or equivalently by improving match efficiency) increase realized output and the measured Solow residual. Hence, *demand shocks can look like technology shocks* in aggregate data.
- **Empirical hook.** Shopping time from the American Time Use Survey (ATUS) is procyclical and positively correlated with both GDP and the Solow residual (Figure 1). The model uses this to identify the “search” (demand) shock separately from true technology and investment-specific shocks.
- **Quantitative result.** In Bayesian estimation, the search-related preference shock explains a large share of the variance of GDP and measured TFP (roughly 40–60% across specifications), and almost all of the variance of shopping time.

2 Section 2: a competitive-search goods market (static baseline)

2.1 Environment

There is a unit mass of potential producers (“firms”/“stores”) and a unit mass of households. The friction is that households must *search* to find a producer and complete a trade. A “submarket” is indexed by a price and a congestion level (market tightness). Competitive search means:

- Firms *post* a market bundle (price and implied tightness).
- Households *direct* their search toward the market that maximizes their expected utility.
- Free entry/competition pins down the market bundle so that the allocation is efficient (welfare theorems).

2.2 Matching technology

Let D denote aggregate search effort (number of searching customers) and T the measure of producing locations in a market. Matches are produced by a Cobb–Douglas matching function

$$M = \mu D^\phi T^{1-\phi}, \quad 0 < \phi < 1, \mu > 0, \quad (1)$$

so that with market tightness $q \equiv D/T$,

$$\Psi_d(q) = \frac{M}{D} = \mu q^{\phi-1}, \quad \Psi_T(q) = \frac{M}{T} = \mu q^\phi. \quad (2)$$

Interpretation:

- $\Psi_d(q)$ is the probability a searcher successfully finds a seller (decreasing in q).
- $\Psi_T(q)$ is the expected number of customers per seller (increasing in q).

2.3 Static equilibrium intuition (why only one active market)

Because households and firms are ex ante identical, in equilibrium everyone chooses the same submarket. Competitive search yields a unique active market that implements the planner allocation. Intuitively, if two submarkets were active, households would arbitrage across them until equalized, and firms would be indifferent; the concavity and monotonicity of Ψ_d, Ψ_T make this outcome collapse to a single market.

2.4 Proposition 1 (existence/uniqueness and efficiency): statement and proof sketch

Proposition (paraphrase). In the endowment version of the model, there is a unique competitive-search equilibrium with a unique active market. The allocation coincides with the planner solution.

Proof sketch (economic). Competitive search equilibria in this environment satisfy a welfare theorem: because households direct search and firms post contracts, the equilibrium contract maximizes household utility subject to firms breaking even. This is isomorphic to a planner choosing (D, C) subject to the matching technology. Concavity of the matching function in D ensures a unique optimum. Hence the equilibrium is unique and efficient.

3 Section 2.1: introducing shocks and dynamics in a tractable production economy

This section is the paper’s “workhorse intuition” for why a demand shock can move measured TFP.

3.1 Production with a product-market matching friction

There is a unit measure of firms. Each firm transforms labor input n into a good:

$$f = z n^\alpha, \quad 0 < \alpha < 1, \quad (3)$$

where z is *true* productivity.

Only matched output becomes realized GDP/consumption. Using (1)–(2) with $T = 1$, realized output is

$$Y = M \cdot \frac{f}{T} = \Psi_T(D) z N^\alpha = \mu z D^\phi N^\alpha. \quad (4)$$

3.2 Measured TFP as a mismeasured Solow residual

National accounts include labor (and capital in the full model) as inputs, but do *not* include shopping/search time as a factor input. If an econometrician fits the standard production function $Y = AN^\alpha$, the implied measured TFP is

$$A \equiv \frac{Y}{N^\alpha} = \mu z D^\phi. \quad (5)$$

Key point: even when z is constant, fluctuations in D mechanically generate fluctuations in A .

3.3 Two-period dynamics (storage)

To introduce intertemporal choices without capital, goods can be stored at zero interest. With savings s ,

$$c = Y - s, \quad c' = Y' + s. \quad (6)$$

3.4 Preference (demand) shock that operates through search

Preferences are of the GHH form (to remove wealth effects and deliver procyclical search):

$$u(c, d, n; \theta) = \frac{1}{1-\gamma} \left[c - \theta_d \frac{d^{1+1/\eta}}{1+1/\eta} - \chi \theta_n \frac{n^{1+1/\nu}}{1+1/\nu} \right]^{1-\gamma}. \quad (7)$$

Here θ_d raises the marginal disutility of shopping/search. A *positive demand shock* corresponds to *lower* θ_d (shopping becomes easier/more attractive), raising D .

3.5 Proposition 2 (comparative statics and “demand looks like technology”)

The paper derives log-linear expressions showing that shocks to z and to θ_d both move (Y, C, N, S) in the same direction, but they have different implications for D :

- An increase in z raises output directly and also raises the return to searching, inducing higher D .
- A decrease in θ_d raises D directly, which raises Y through (4).

A compact way to summarize the central implication is to log-linearize (5):

$$\hat{A} = \hat{z} + \phi \hat{D}, \quad (8)$$

so any shock that moves $\hat{D}$ moves the measured Solow residual even when true technology does not change.

3.6 Identification logic (why shopping time matters)

Because aggregate observables (Y, C, N, I) respond similarly to θ_d and z , the model needs extra information to disentangle them. Two complementary strategies are used in the paper:

1. **Direct observation of search:** use shopping time (ATUS) as a proxy for D . Then D ’s comovement helps separate θ_d from z .
2. **Time-series restriction:** treat z as a (more persistent) trend shock and θ_d as stationary, so the different persistence helps identification even without shopping time.

4 Section 3: the full stochastic growth model with consumption and investment markets

The quantitative model embeds the product-market search friction in a standard neoclassical growth framework.

4.1 Technology (two sectors and an investment-specific shock)

There is a measure T_c of consumption-good firms and T_i of investment-good firms, with $T_c + T_i = 1$.

Each sector uses the same concave production function $f(k, n) = k^{\alpha_k} n^{\alpha_n}$. Technology shocks:

$$\text{Consumption sector: } F_c = z_c f(k_c, n_c), \quad z_c \equiv z, \quad (9)$$

$$\text{Investment sector: } F_i = z_i f(k_i, n_i), \quad z_i \equiv z z_I, \quad (10)$$

where z is the common (trend) technology component and z_I is an investment-specific technology shock.

4.2 Households: shopping for consumption and investment goods

Households choose consumption c , shopping efforts (d_c, d_i) , labor n , and next-period capital allocations (k'_c, k'_i) . Shopping produces *found* consumption and investment goods:

$$c = d_c \Psi_{d,c}(q_c) F_c, \quad (11)$$

$$i = d_i \Psi_{d,i}(q_i) F_i. \quad (12)$$

Budget constraint (consumption numéraire $p_c = 1$):

$$c + p_i i = \pi + k_c R_c + k_i R_i + nW, \quad (13)$$

capital accumulation:

$$i = k'_c + k'_i - (1 - \delta)(k_c + k_i). \quad (14)$$

Preferences remain GHH in $(c, d_c + d_i, n)$ with shocks $\theta = (\theta_d, \theta_n)$ and a stationary transformation in the presence of growth (the paper scales preference weights by lagged z to ensure stationarity under GHH).

4.3 Firms and competitive search contracts

Firms rent capital and labor in spot markets and post a contract/market bundle (p_j, q_j, F_j) for $j \in \{c, i\}$, where F_j is the quantity placed for sale. Given the household value of shopping in each market, the firm chooses $(p_j, q_j, F_j, k_j, n_j)$ to maximize expected profits subject to households being willing to visit its market (an incentive/participation constraint). Because the firm's problem is static (no dynamic state at the firm level), equilibrium can be solved via the planner.

4.4 Planner problem and key optimality conditions

Let K_j, N_j, D_j denote aggregate capital, labor, and search in sector j . The planner chooses $(D_c, D_i, N_c, N_i, K'_c, K'_i)$ to maximize utility subject to resource and matching constraints. The crucial intratemporal condition for consumption search equates the marginal rate of substitution between search and consumption to the marginal product of search through matching:

$$-\frac{u_D}{u_C} = \frac{\partial}{\partial D_c} \left(\Psi_{T,c}(D_c) z_c f(K_c, N_c) \right), \quad (15)$$

and analogously for investment search (with the investment price p_i appearing as a shadow value).

The model-implied relative price of investment is

$$p_i = \frac{\Psi_{T,c}(D_c) z_c \partial f(K_c, N_c) / \partial N_c}{\Psi_{T,i}(D_i) z_i \partial f(K_i, N_i) / \partial N_i}. \quad (16)$$

Intuition: p_i falls when z_I rises (investment goods become cheaper to produce) and rises when the consumption market is relatively tighter than the investment market.

The Euler equation resembles the standard one but incorporates matching in the marginal return to capital, because future output depends on the probability goods are found.

5 Section 3.5: cross-sectional discipline and identification of (ϕ, η)

The model yields sharp cross-sectional implications because households with higher expenditure choose different shopping intensity and face different prices in equilibrium.

5.1 Equilibrium price schedule and shopping intensity

In the cross-sectional extension (households differ in expenditure y_j), competitive search implies a price schedule $p_j = p(y_j)$ and search choice $D_j = D(y_j)$.

A key implication is that the cross-sectional elasticity of prices with respect to expenditure is pinned down by (ϕ, η) :

$$\frac{\text{sd}(\log p_j)}{\text{sd}(\log y_j)} = \frac{\phi}{1 + \eta(1 - \phi)}. \quad (17)$$

Similarly, the expenditure elasticity of shopping/search is

$$\frac{\partial \log D_j}{\partial \log y_j} = \frac{\eta(1 - \phi)}{1 + \eta(1 - \phi)}. \quad (18)$$

These moments are exactly what Table 1 uses to calibrate ϕ and η .

6 Section 5: Bayesian estimation and what the main tables show

6.1 Estimated shocks

The estimation uses four shocks:

- $\theta_{d,t}$: search disutility / product-market friction shock (“demand/search shock”).
- $\theta_{n,t}$: labor disutility shock (moves labor wedge).
- g_t : the (trend) growth component in z_t (technology shock).
- $z_{I,t}$: investment-specific technology shock (moves relative price of investment).

Each follows an AR(1):

$$x_t = \rho_x x_{t-1} + \sigma_x \varepsilon_{x,t}, \quad \varepsilon_{x,t} \sim \mathcal{N}(0, 1),$$

with independent innovations.

6.2 Data moments and mixed frequency

Targets:

- quarterly GDP growth,
- quarterly investment growth,
- quarterly labor productivity growth,
- and either (i) annual shopping time (ATUS, 2003–2019) *or* (ii) the quality-adjusted relative price of investment (long sample).

A mixed-frequency state-space representation is used when shopping time is included.

7 Figure and Tables

7.1 Figure 1: shopping time is procyclical and correlated with measured TFP

- Figure 1 plots detrended shopping time, GDP, and the Solow residual (TFP) from 2003–2019.
- Shopping time is **procyclical**: it falls sharply in the Great Recession and rises in expansions.
- Shopping time is positively correlated with GDP and with the Solow residual (the figure reports correlations around 0.5).
- **Interpretation:** If shopping/search is an unmeasured input that scales realized output, then procyclical shopping time mechanically generates procyclical measured TFP. This is the empirical motivation for the model and for using shopping time to identify the search shock.

7.2 Table 1: calibration and how the “search” parameters are pinned down

Table 1 splits parameters into (i) standard macro parameters (risk aversion, discounting, production shares, depreciation) and (ii) search-economy parameters.

- The key search parameters are **matching elasticity** ϕ and **shopping elasticity** η .
- They are disciplined by cross-sectional moments: price dispersion relative to expenditure dispersion (17), and the shopping-time expenditure elasticity (18).
- The calibration also normalizes the steady state by targeting average hours, utilization, and consumption/investment shares.

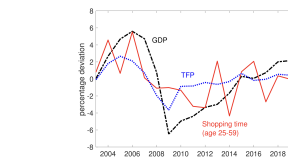


FIGURE 1 Shopping time versus GDP and TFP
Note: The figure plots average shopping time for the working age population (25–59) against GDP and the Solow residual. All series are linearly detrended. Shopping time is measured as the average time spent shopping for goods, professional and personal care services, and household services, plus travel time associated with this shopping. Data source: ATUS from U.S. Bureau of Labor Statistics (2021) and NIPA from U.S. Bureau of Economic Analysis (2020).

Targets	Value	Parameter	Calibration
First group: parameters set exogenously			
Risk aversion	1	γ	1
Real interest rate	4%	β	0.997
Average growth rate	3%	$\bar{g}$	0.74%
Frisch elasticity	0.72	ν	0.72
Second group: normalizations			
Steady-state output	1	$E(z_t)$	1.22
Relative price of investment	30%	z	7.99
Fraction of time spent working	81%	u	0.86
Capacity utilization	80	T_z	0.80
Consumption share of output			
Third group: standard targets			
Investment share of output	0.20	δ	1.08%
Physical capital-to-output ratio	2.75	a_k	0.23
Labour share of output	0.67	α_n	0.45
Fourth group: targets specific to this economy			
Ratio cross-sectional st. dev. of goods prices to st. dev. of cons. expenditure	26.6%	ϕ	0.32
Shopping time expenditure elasticity	7.5%	η	0.20

Parameter	Density	Post(1)	Post(2)	Mean	90% Int.
Prices and posteriors for the shock parameters likelihood = 603.6					
ρ_d	Beta	0.900	0.05	0.879	[0.808, 0.951]
ρ_g	Beta	0.100	0.05	0.205	[0.046, 0.330]
ρ_z	Beta	0.900	0.05	0.874	[0.793, 0.996]
ρ_n	Beta	0.900	0.05	0.968	[0.946, 0.992]
σ_d	Inv Gamma	0.010	Inf	0.097	[0.041, 0.113]
σ_g	Inv Gamma	0.010	Inf	0.008	[0.006, 0.009]
σ_z	Inv Gamma	0.010	Inf	0.012	[0.009, 0.014]
σ_n	Inv Gamma	0.010	Inf	0.016	[0.014, 0.018]
Variance decomposition					
	θ_L	g	z_I	θ_n	
Y	38.15	44.64	5.45	11.77	
Y/N	23.18	40.79	6.86	29.17	
Investment	21.07	11.50	62.10	5.32	
Investment price	19.44	6.38	76.13	4.04	
TFP	38.54	48.80	8.70	5.77	
Consumption	28.31	41.48	20.70	9.51	
Labour	10.15	23.53	0.87	65.76	
Shopping	98.56	1.25	0.03	0.16	

Notes: The estimation targets the data series of quarterly GDP growth, labour productivity growth, investment growth, and annual shopping time. U.S. data, 2003–19.

7.3 Table 2: benchmark estimation with shopping data (short sample, mixed frequency)

Upper panel (posteriors). The estimated persistence and volatilities show:

- ρ_n (labor disutility) is very persistent,
- ρ_d (search disutility) is persistent but less so,
- g is relatively low persistence in the short sample,
- z_I is persistent and has a small innovation variance.

Lower panel (variance decomposition).

- The search shock θ_d explains a **large share of GDP and measured TFP** (roughly 40% in the benchmark).
- True technology (g) also explains a large share.
- The investment-specific shock dominates the variance of the relative investment price and contributes importantly to investment fluctuations.
- Shopping time variance is **almost entirely** explained by θ_d , consistent with the model's identification strategy.

7.4 Table 3: benchmark estimation without shopping data (longer sample, using P_i)

Dropping shopping time and using the investment-price series shifts identification to persistence restrictions (trend vs stationary).

- The estimated θ_d shock becomes even more important for GDP and TFP variance in this specification.
- Investment-price movements are explained mostly by z_I , as in standard IST-shock narratives, but θ_d can also load on P_i through general equilibrium (because tighter consumption markets affect the shadow value of investment).

TABLE 3
Bayesian estimation: benchmark model with P_i , no shopping data

Priors and posteriors for the shock parameters likelihood = 2404.8					
Parameter	Density	Para(1)	Para(2)	Mean	90% Intv.
ρ_d	Beta	0.900	0.05	0.926	[0.915, 0.937]
ρ_g	Beta	0.100	0.05	0.603	[0.597, 0.607]
ρ_I	Beta	0.900	0.05	0.954	[0.945, 0.961]
ρ_n	Beta	0.900	0.05	0.998	[0.997, 0.9997]
σ_d	Inv Gamma	0.010	Inf	0.241	[0.222, 0.259]
σ_g	Inv Gamma	0.010	Inf	0.011	[0.010, 0.012]
σ_I	Inv Gamma	0.010	Inf	0.014	[0.013, 0.016]
σ_n	Inv Gamma	0.010	Inf	0.017	[0.015, 0.018]

Variance decomposition				
	θ_d	g	z_I	θ_n
Y	66.10	28.09	2.16	3.64
Y/N	41.95	45.75	3.03	9.27
Investment	55.05	8.97	34.47	1.50
Investment price	47.83	0.09	51.43	0.65
TFP	58.03	37.92	3.02	1.02
Consumption	53.75	35.03	7.50	3.72
Labour	36.85	19.68	0.39	43.08
Shopping	99.68	0.29	0.01	0.03

Notes: The estimation targets the data series of quarterly GDP growth, labour productivity growth, investment growth, and relative price of investment data, 1967 to 2019.

TABLE 4
Robustness: alternative preference shocks

	LML	Δ LML	$std(\theta_d)$	$std(\theta_n)$	$std(\theta_c)$	$std(\beta)$	$std(\bar{n})$	$corr(\hat{\gamma}, \hat{\gamma} - \bar{n})$
Use shopping time data 2003–19								
GHH preferences								
(1) θ_d benchmark	603.6	−6.5	0.097	0.016	–	–	1.05	0.50
(2) θ_c	485.8	−124.3	–	0.091	0.087	–	7.19	−0.92
(3) β	78.1	−532.0	–	0.095	–	0.020	8.97	0.79
Additively separable preferences								
(4) θ_d	610.1	0	0.104	0.013	–	–	0.69	0.76
(5) θ_c	470.8	−139.3	–	0.096	0.100	–	5.05	−0.84
(6) β	223.4	−386.8	–	0.081	–	0.027	7.70	0.34
Drop shopping time data, 1967–2019								
GHH preferences								
(7) θ_d benchmark	2404.8	−196.9	0.241	0.017	–	–	1.34	0.80
(8) θ_c	2257.1	−344.7	–	0.058	0.071	–	5.16	−0.83
(9) β	2510.1	−91.6	–	0.018	–	0.001	1.07	0.32
Additively separable preferences								
(10) θ_d	2601.8	0	0.141	0.018	–	–	0.91	0.64
(11) θ_c	2513.0	−88.7	–	0.020	0.019	–	1.24	0.27
(12) β	2561.4	−40.3	–	0.017	–	0.001	1.06	0.35
U.S. data, 2003–19 quarterly							0.65	0.56

Notes: The table documents the estimation results for alternative specifications of preference shocks. LML denotes the log marginal likelihood and Δ LML denotes the log marginal likelihood differentials (or log Bayes factors). For each data set, Δ LML is calculated relative to the specification with the highest marginal likelihood. Rows (1) and (7) refer to the two benchmark economies with a θ_d shock (cf. Tables 2 and 3).

7.5 Table 4: alternative preference shocks (why θ_d is special)

This table compares the benchmark search-disutility shock to other preference shocks (e.g., shocks to consumption weight or discount factor), and to alternative preference specifications.

- Models where the main preference shock is *not* the search disutility fit the data much worse (large drops in log marginal likelihood).
- The search shock is special because it creates *comovement in GDP, consumption, and measured TFP* via the utilization/matching channel, while other demand shocks often imply counterfactual behavior (e.g., countercyclical productivity).

7.6 Table 5: robustness (restrictions, alternative data, alternative preferences)

Across many robustness checks (imposing moment restrictions, estimating in levels, separable preferences, dropping shopping time), the qualitative message remains:

- θ_d continues to explain a sizable share of GDP and TFP fluctuations.
- The magnitude varies (roughly 40–60%) but does not collapse.

TABLE 5
Further robustness analysis

	ρ	η	LML	$std(\theta_d)$	θ_d	g	z_I	θ_d	g	z_I
Use shopping time data 2003–19, mixed frequency										
(1) Benchmark: impose (34)–(35)	0.32	0.20	603.6	0.097	38.15	44.64	5.45	38.74	48.80	8.70
(2) Alternative: $\eta = \nu$	0.41	0.72	588.1	0.033	40.31	46.82	4.02	42.04	48.64	6.59
(3) Impose only equation (34)	0.30	0.09	606.3	0.221	41.79	39.59	6.45	41.54	44.46	10.26
(4) Search only for c	0.32	0.20	609.3	0.096	29.22	41.09	17.39	25.90	44.81	27.22
(5) Estimation in levels	0.32	0.20	592.0	0.120	31.85	39.45	5.51	31.69	39.10	8.21
(6) Additively separable preferences	0.32	0.20	610.1	0.104	39.00	41.28	13.10	41.27	51.14	5.72
Drop shopping time data, 1967–2019, quarterly, GHGHC, d, n										
(7) Benchmark: impose (34)–(35)	0.32	0.20	2484.8	0.241	66.10	28.09	2.16	58.03	37.92	3.02
(8) Alternative: $\eta = \nu$	0.41	0.72	2349.0	0.083	64.02	32.27	1.24	58.04	39.51	1.77
(9) Impose only equation (34)	0.30	0.09	2407.9	0.352	69.29	24.79	2.38	61.13	34.50	3.39
Drop shopping time data, 2003–19 quarterly, GHGHC, d, n										
(10) Benchmark: impose (34)–(35)	0.32	0.20	767.8	0.206	69.45	26.26	1.88	61.56	35.09	2.67

Notes: The table documents the estimation results for under alternative calibrations and data choices. All estimations target the quarterly U.S. data series of GDP growth, labour productivity growth, and investment growth. Rows (1)–(6) target aggregate ATUS shopping time, while rows (7)–(10) target the real price of investment goods. LML denotes the log marginal likelihood. The benchmark model (rows 1 and 7) has GHGHC, d, n preferences and uses the benchmark calibration in Table 1, imposing conditions (34)–(35). Row (2) uses the alternative calibration in Table 1, assuming $\eta = \nu$ and imposing equation (34). Row (3) imposes only equation (34) and estimates ρ . Row (4) assumes there is a search friction only for consumption goods. Row (5) estimates the benchmark model in levels (as opposed to growth rates). Row (6) assumes additively separable preferences. Rows (7)–(9) correspond to rows (1)–(3), using investment price data instead of the series for shopping time. Row (10) is the same as row (7) but uses a shorter time sample.

TABLE 6
Search-effort amplification

Measure 1: Instantaneous impact on TFP and GDP					
	TFP		Output		
	Search	Fixed search	Search	Fixed search	
g	1.075	1.000	1.339	1.259	
z_I	0.212	0.203	0.234	0.237	
θ_d	-0.044	-0.000	-0.254	-0.239	
θ_d	-0.058	0.000	-0.074	0.000	
Measure 2: Variance from each shock alone					
	Variance of TFP			Variance of Output	
	Search	Fixed search	% increase	Search	Fixed search
g	0.428	0.359	19.1	0.625	0.546
z_I	0.064	0.060	6.4	0.076	0.080
θ_d	0.0005	0.000	-	0.165	0.146
θ_d	0.341	0.000	-	0.534	0.000
Total	0.834	0.420	98.8	1.400	0.772

Notes: The upper panel reports the contemporaneous elasticities of TFP and GDP to the shocks z_I , z_I , and θ_d in economies with endogenous search ($\eta > 0$) and with fixed search ($\eta = 0$), respectively. The lower panel reports the variance of TFP and output attributable to each shock in the benchmark economy (“Search”) and in and the fixed-search economy. The “% increase” column reports how much the variance contribution of each shock is amplified when going from constant search to endogenous search effort.

TABLE 7
Regression estimation: with capacity utilization on capital

Priors and posteriors for the shock parameters likelihood = 610.7					
Parameter	Density	Para(1)	Para(2)	Mean	90% Intv.
β_d	Beta	0.900	0.05	0.863	[0.794, 0.940]
β_g	Beta	0.100	0.05	0.190	[0.100, 0.294]
β_n	Beta	0.900	0.05	0.973	[0.932, 0.997]
β_o	Beta	0.900	0.05	0.974	[0.956, 0.994]
σ_d	Inv Gamma	0.010	Inf	0.006	[0.079, 0.111]
σ_g	Inv Gamma	0.010	Inf	0.006	[0.004, 0.007]
σ_n	Inv Gamma	0.010	Inf	0.010	[0.007, 0.014]
σ_o	Inv Gamma	0.010	Inf	0.014	[0.011, 0.016]
σ_η	Inv Gamma	0.100	Inf	0.279	[0.035, 0.549]
Var Decomposition					
	θ_d	g	z_I	θ_n	
Y	45.35	30.68	13.44	10.53	
Y/N	37.07	30.46	18.11	14.36	
Investment	34.53	7.84	51.88	5.75	
Investment price	3.46	0.36	95.68	0.51	
TFP	47.75	32.63	19.09	0.53	
Consumption	26.97	51.71	11.49	9.83	
Labour	16.53	19.08	1.91	62.47	
Capital intensity	84.96	4.78	8.77	1.49	

Notes: The estimation targets the data series of GDP, labour productivity, investment, and shopping time. Quarterly U.S. data, 2003:1 to 2019:IV. The inverse of σ_η is the elasticity of capital depreciation rate to capital usage.

7.7 Table 6: endogenous search as an amplification mechanism

This is the key propagation result.

- **Instantaneous elasticities:** with endogenous search, a technology shock g has a larger impact on GDP/TFP than in a fixed-search economy, because z increases the return to searching and households endogenously raise D .
- **Variance amplification:** allowing D to respond endogenously raises the variance of both GDP and measured TFP attributable to non-search shocks (notably the technology shock), and it adds a new source of TFP variation from θ_d itself.

7.8 Table 7: allowing for capital utilization does not kill the mechanism

A standard DSGE channel is variable capital utilization. The paper extends the model with utilization h and cost

$$\psi(h) = \xi \frac{h^{1+\sigma_a} - 1}{1 + \sigma_a}.$$

Even with this additional “unmeasured input” margin:

- θ_d still explains a large share of GDP and TFP volatility.
- Hence, the product-market search channel is not just a relabeling of capital utilization.

7.9 Tables 8–10: storable goods and goods/services decomposition

These tables extend the model to allow:

- separate goods and services sectors,
- inventories (goods can be stored and traded), and

- CES aggregation:

$$C = \left[\omega_c M_c^{\frac{\zeta-1}{\zeta}} + (1 - \omega_c) S_c^{\frac{\zeta-1}{\zeta}} \right]^{\frac{\zeta}{\zeta-1}}, \quad I = \left[\omega_i M_i^{\frac{\zeta-1}{\zeta}} + (1 - \omega_i) S_i^{\frac{\zeta-1}{\zeta}} \right]^{\frac{\zeta}{\zeta-1}}.$$

The concern is that if goods can be stored, low demand today may not imply permanently lost output (it becomes inventory), potentially weakening the mechanism. The results show robustness:

- θ_d remains an important driver of GDP and especially measured TFP.
- Inventory dynamics (inventory-to-sales) are strongly tied to θ_d , which is consistent with the idea that low demand leaves more goods unsold and thus in inventory.

TABLE 8
Bayesian estimation: model with storable goods, shopping time, short sample

Priors and posteriors for the shock parameters likelihood = 614.2

Parameter	Density	Para(1)	Para(2)	Mean	99% Inv.
ρ_d	Beta	0.900	0.05	0.855	[0.771, 0.937]
ρ_s	Beta	0.100	0.05	0.230	[0.106, 0.352]
ρ_i	Beta	0.900	0.05	0.901	[0.823, 0.989]
ρ_n	Beta	0.900	0.05	0.967	[0.946, 0.991]
σ_d	Inv Gamma	0.010	Inf	0.097	[0.081, 0.113]
σ_s	Inv Gamma	0.010	Inf	0.007	[0.006, 0.008]
σ_i	Inv Gamma	0.010	Inf	0.014	[0.011, 0.016]
σ_n	Inv Gamma	0.010	Inf	0.015	[0.013, 0.018]

	Variance decomposition			
	θ_d	g	z_t	θ_n
Y	34.57	38.73	10.24	16.47
Y/N	17.76	39.34	12.96	29.94
Investment	8.10	6.60	79.27	6.03
Investment price	1.11	1.75	94.90	2.24
TFP	45.38	23.64	16.12	14.87
Consumption	23.38	39.36	16.09	12.17
Labour	9.08	20.77	0.66	69.49
Service	43.60	40.87	1.91	13.62
Goods	3.54	16.44	65.61	14.41
Service price	24.82	62.43	11.12	1.64
goods/service	18.11	9.32	71.15	1.42
inventory/sale	78.38	5.10	13.73	2.79

Notes: The table shows the Bayesian estimation for the shopping model with storable goods and services. The estimation targets the data series of output, labour productivity, investment, and shopping time. The sample is quarterly U.S. data, from Q1 in 2003 to Q4 in 2019.

TABLE 9
Bayesian estimation: model with storable goods, long sample

Priors and posteriors for the shock parameters likelihood = 2599.3

Parameter	Density	Para(1)	Para(2)	Mean	99% Inv.
ρ_d	Beta	0.900	0.05	0.899	[0.895, 0.998]
ρ_s	Beta	0.100	0.05	0.020	[0.0068, 0.0407]
ρ_i	Beta	0.900	0.05	0.632	[0.5907, 0.6725]
ρ_n	Beta	0.900	0.05	0.990	[0.9834, 0.9970]
σ_d	Inv Gamma	0.010	Inf	0.212	[0.1930, 0.2297]
σ_s	Inv Gamma	0.010	Inf	0.017	[0.0151, 0.0180]
σ_i	Inv Gamma	0.010	Inf	0.023	[0.0199, 0.0257]
σ_n	Inv Gamma	0.010	Inf	0.024	[0.0222, 0.0262]

	Variance decomposition			
	θ_d	g	z_t	θ_n
Y	32.91	50.87	7.47	8.74
Y/N	22.11	47.43	10.67	19.79
Investment	4.94	13.77	77.70	3.58
PI	13.52	7.63	77.05	1.80
TFP	47.63	27.63	15.36	9.38
Consumption	37.24	46.69	8.99	7.08
Labour	10.59	40.39	0.72	48.29
Service	40.53	51.80	1.47	7.00
Goods	2.72	27.28	60.05	9.95
Service price	20.31	74.89	4.28	0.52
goods/service	28.73	13.92	56.75	0.60
inventory/sale	99.80	0.05	0.12	0.03

Notes: The table shows the Bayesian estimation for the shopping model with storable goods and services. The estimation targets the data series of output, labour productivity, investment, and investment price. The sample is quarterly U.S. data, from Q1 in 1967 to Q4 in 2019.

TABLE 10
Robustness analysis—goods and services

	Variance decomposition Y				Variance decomposition TFP				
	Lhood	std(θ_d)	θ_d	g	z_t	θ_d	g	z_t	Rel. std goods
Use shopping time data 2003–19, mixed frequency									
(1) Benchmark:									
non-durable	614.2	0.097	34.57	38.73	10.24	45.38	23.64	16.12	1.29
(2) Alternative: durable	601.8	0.098	32.85	35.62	15.84	42.00	20.77	23.85	1.45
(3) Goods no frictions, non-durable	604.5	0.099	32.09	42.08	11.11	43.63	24.94	17.40	1.31
(4) Goods no frictions, durable	609.8	0.098	31.94	41.74	10.37	42.97	25.46	17.15	1.30

Notes: The table documents the estimation results for under alternative calibrations and model assumptions with both goods and services. All estimations target the quarterly U.S. data series of GDP growth, labour productivity growth, investment growth, and aggregate shopping time. Row (1) and (2) assume non-durable goods, row (2) and (4) have durable goods with depreciation of 10% annually. Row (3) and (4) assume that goods are not subject to search friction, but services are.

8 Short summary

1. **Accounting identity:** realized output equals capacity times a matching/utilization term. The utilization term depends on customer search.
2. **Measurement:** search effort is an input but is not counted in standard input measures, so it shows up in the Solow residual.
3. **Implication:** some “TFP” fluctuations are really product-market tightness fluctuations (demand/search), not true technology.
4. **Quantitative importance:** estimated search shocks explain a large fraction of GDP and measured TFP volatility; shopping time provides a clean identifying moment.

9 Conclusion

9.1 Summary

The paper’s central conclusion is that **aggregate demand shocks can look like technology shocks** once goods-market **matching/search frictions** are explicitly modeled. When households search more intensively for goods, more producers successfully match with customers, so the economy produces *more output with the same installed capital and labor*. If an econometrician (or a standard neoclassical production function) ignores household search effort as an input, this additional output appears as a rise in the **Solow residual** (measured TFP). Hence, **demand-driven**

fluctuations can generate procyclical measured productivity even with flexible prices and otherwise standard neoclassical structure.

Quantitatively, the paper estimates a DSGE model with a product-market search friction using Bayesian methods. Business cycles are driven by **two preference shocks** (shopping/search and labor supply) and **two technology shocks** (neutral technology growth and investment-specific technology). The main empirical conclusion is that **preference shocks affecting search effort are a major driver of U.S. business-cycle fluctuations** in GDP, consumption, and measured TFP, and that this result is robust across extensions (capital utilization; inventories/durables).

9.2 Key quantitative conclusions (the “headline numbers”)

The paper’s variance decompositions imply that the **search-demand shock** (denoted θ_d) accounts for a large share of fluctuations in output and measured TFP:

- **Benchmark estimation targeting shopping time (ATUS, 2003–2019):** θ_d explains about **38%** of GDP variance and about **39%** of measured TFP variance (Table 5, row (1): 38.15% for GDP; 38.74% for TFP).
- **Estimation not targeting shopping time (using investment-price data):** θ_d explains roughly **66–69%** of GDP variance and **58–62%** of TFP variance (Table 5, rows (7)–(10): GDP 66.10–69.45%, TFP 58.03–61.56%).
- **Amplification channel (endogenous search amplifies other shocks):** compared to a “fixed search” economy, endogenous search raises the instantaneous passthrough of the neutral technology-growth shock to TFP from 1.000 to 1.075 (about **+8%**), and increases the variance of TFP attributable to that shock by about **19.1%** (Table 6).

Interpretation: even under conservative identification (explicitly disciplining the model with observed shopping time), demand/search shocks account for a *technology-like* fraction of measured TFP fluctuations.

9.3 Economic intuition: why demand shocks become “technology shocks”

A single logic ties the mechanism together:

1. **Matching friction creates idle productive capacity.** Because producers do not automatically meet customers, some potential production is unrealized each period.
2. **Household search is productive.** More shopping/search effort increases the probability of matching with sellers, raising realized output even if measured inputs (capital, labor) are unchanged.
3. **Measured TFP rises when search is omitted from the production function.** If the accounting framework treats output as $Y = AF(K, N)$ but the true economy is closer to $Y = \tilde{A}F(K, N, \text{search})$, then cyclical changes in search show up mechanically as cyclical changes in A (the Solow residual).
4. **Therefore, demand shocks can mimic productivity shocks in impulse responses.** A positive demand/search shock raises output, consumption, investment, and measured TFP in ways that resemble a technology shock, but it does so through higher match intensity rather than higher true production efficiency. Shopping time (or other measures of search effort) is therefore a key observable to separate the shocks.

9.4 Broader implication for macro modelling

The paper’s bottom line for macro practice is:

If measured TFP is strongly affected by demand-induced changes in search/matching intensity, then calibrating or estimating business-cycle models with “technology shocks” may be capturing demand shocks instead.

This provides a structural rationale for why demand-side forces can generate “productivity-like” fluctuations even under flexible prices, and it suggests that demand shocks can substitute for (some) productivity shocks in New Keynesian and search-and-matching macro traditions.

9.5 Takeaway

With product-market search frictions, higher demand induces more search, which raises matches and output, so demand shocks generate procyclical measured TFP and account for a large share of GDP and Solow-residual fluctuations.